TECHNICAL REPORT

EMC TEST REPORT

Report No:	MK25030212P03-E01
Report Date:	2025-4-10
Applicant:	Shenzhen HonZen Vision Technology Co., Ltd.
Product:	Sports camera
Brand Name:	N/A
Model No:	H88, i3, i3s
Test Standards:	ETSI EN 301 489-1 V2.2.3(2019-11) ETSI EN 301 489-17 V3.3.1 (2024-09)
Test result:	The EMC testing has been performed on the submitted samples and found in compliance with council EMC Directive 2014/30/EU and RE Directive 2014/53/EU

Approved By

Vivian Jiang

EMC Manager

Dated: April 10, 2025

Results appearing herein relate only to the sample tested

The technical reports is issued errors and omissions exempt and is subject to withdrawal at

TMC Testing Services (Shenzhen) Co., Ltd.

1st Floor, Block A1, Zone A, Xinshidai Gongrong Industrial Park, No. 2, Shihuan Road, Shiyan Street, Baoan District, Shenzhen, China

Tel:86-755-86642861 Email: cert@tmc-lab.com

1. General Information

1.1 Notes

The test results of this report relate exclusively to the test item specified in 1.5. The TMC Lab does not assume Responsibility for any conclusions and generalizations drawn from the test results with regard to other specimens or samples of the type of the equipment represented by the test item. The test report may only be reproduced or published in full. Reproduction or publication of extracts from the report requires the prior written approval of the TMC Lab.

1.2 Testing Laboratory

TMC Testing Services (Shenzhen) Co., Ltd.

1st Floor, Block A1, Zone A, Xinshidai Gongrong Industrial Park, No. 2, Shihuan Road, Shiyan Street, Baoan District, Shenzhen, China

Tel: 86-755-86642861 Email: cert@tmc-lab.com

Internet: www.tmc-lab.com

1.3 Details of Applicant

Name: Shenzhen HonZen Vision Technology Co., Ltd.

Address:4/F Building B, Yili Science Park, Guanhu Street, Longhua District, Shenzhen

1. 4 Application Details

Date of Receipt of Application: April 2, 2025

Date of Receipt of Test Item: April 2, 2025

Date of Test: April 2, 2025-April 10, 2025

1.5 Test Item

Manufacturer: Shenzhen HonZen Vision Technology Co., Ltd.

Address: 4/F Building B, Yili Science Park, Guanhu Street, Longhua District, Shenzhen

Brand Name: N/A

Model No.: H88

Additional Model No.: i3, i3s

Additional Brand Name: N/A

Description: Sports camera

Additional Information

Frequency: Wi-Fi 2412-2472 MHz
Antenna type: FPC Antenna
Type of Modulation: BPSK, QPSK,16QAM,64QAM, OFDM
Ant. Gain: 2.62 dBi
Hardware Version:/
Software Version: /
Extreme Temp. Tolerance: -20℃ to +60℃

1.6 List of Ports

Port	Description	Classification[1]	Maximum cable Length	Cable Type
USB Port	-	-	-	-
Line in Port	-	-	-	-

Note [1]prots shall be classified as ac power, dc power or signal/control port.
[2]Maximum cable length corresponding to the appropriate ports shall be classified as ≤3m or ＞3m.

1. 7 Ancillary and Peripheral Devices

Description	Designation	Serial No.	Manufacturer
N/A	-	-	-

1.8 Test Standards

ETSI EN 301 489-1 V2.2.3(2019-11)
Electromagnetic compatibility and Radio spectrum Matters (ERM);
Electromagnetic Compatibility (EMC) standard for radio equipment and services;
Part 1: Common technical requirements
ETSI EN 301 489-17 V3.3.1 (2024-09)
Electromagnetic compatibility and Radio spectrum Matters (ERM);
Electromagnetic Compatibility (EMC) standard for radio equipment and services;
Part 17: Specific conditions for 2.4GHz wideband transmission systems and 5GHz high performance RLAN equipment

Note: All radiated measurements were made in all three orthogonal planes. The values reported are the maximum values.

1.9. Measurement Uncertainty

Item	MU	Remark
Uncertainty for Power point Conducted Emissions Test	2.42dB	
Uncertainty for Radiation Emission test in 3m chamber (30MHz to 1GHz)	3.54dB	Polarize: V
	4.1dB	Polarize: H
Uncertainty for Radiation Emission test in 3m chamber (1GHz to 25GHz)	2.08dB	Polarize: H
	2.56dB	Polarize: V
Uncertainty for radio frequency	0.01ppm	
Uncertainty for conducted RF Power	0.65dB	
Uncertainty for temperature	0.2℃	
Uncertainty for humidity	1%	
Uncertainty for DC and low frequencyvoltages	0.06%	

1.10. Description Of Test Modes

TM1: Operate in WIFI mode.

TM2: charging mode.

TM3: standby mode.

1.11 Test or Witness Test Engineering

Test By: Nina Wu

Printing Name :Nina Wu

2. Technical Test

2. 1 Summary of Test Results

Rule	Description of Test Items	Result
§7.1	Reference to clauses EN 301 489-1 §8.4 AC mains power input/output ports	Compliant
§7.1	Reference to clauses EN 301 489-1§8.3 DC power input/output ports	N/A*
§7.1	Reference to clauses EN 301 489-1 §8.2 Enclosure of ancillary equipment measured on a stand alone basis	Compliant
§7.1	Reference to clauses EN 301 489-1 §8.5 Harmonic current emissions (AC mains input port)	N/A*
§7.1	Reference to clauses EN 301 489-1 §8.6 Voltage fluctuations and flicker (AC mains input port)	N/A*
§7.1	Reference to clauses EN 301 489-1§8.7 Telecommunication ports	N/A*
§7.2	Reference to clauses EN 301 489-1 §9.3 Electrostatic discharge (EN 61000-4-2)	Compliant
§7.2	Reference to clauses EN 301 489-1 §9.2 Radio frequency electromagnetic field (80 MHz to 6 000 MHz)(EN 61000-4-3)	Compliant
§7.2	Reference to clauses EN 301 489-1§9.4 Fast transients, common mode (EN 61000-4-4)	N/A*
§7.2	Reference to clauses EN 301 489-1§9.8 Surges (EN 61000-4-5)	N/A*
§7.2	Reference to clauses EN 301 489-1§9.5 Radio frequency, common mode (EN 61000-4-6)	N/A*
§7.2	Reference to clauses EN 301 489-1 §9.6 Transients and surges in the vehicular environment (ISO 7637-2)	N/A*
§7.2	Reference to clauses EN 301 489-1§9.7 Voltage dips and interruptions (EN 61000-4-11)	N/A*

NOTE：N/A=Not applicable

1. The N/A test items applies to AC powered only. The product is battery powered, without equipped with a power adapter.

2. For EN 301 489-1 §9.6, only those devices used in vehicular environment applies.

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3. LINE CONDUCTED EMISSION

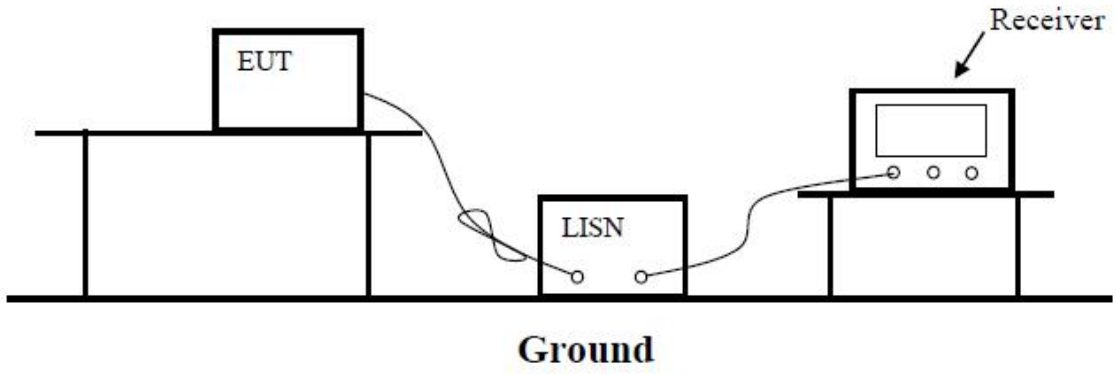
3.1. Conducted Emission Limit

ETSI EN 301 489-1 V2.2.3(2019-11)Clause 8.4.3/EN 55032 Class B

Limits for Line Conducted Emission

Frequency (MHz)	Limit (dB V)	
	Quasi-peak Level	Average Level
0.15 ~ 0.50	66.0 ~ 56.0 *	56.0 ~ 46.0 *
0.50 ~ 5.00	56.0	46.0
5.00 ~ 30.00	60.0	50.0
NOTE1-The lower limit shall apply at the transition frequencies. NOTE2-The limit decreases linearly with the logarithm of the frequency in the range0.15MHz to 0.50MHz.		

3.2. Test Configuration



The setup of EUT is according with per ETSI EN 301 489-1 measurement procedure. The specification used was with the ETSI EN 301 489-1 limits.
The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle. The spacing between the peripherals was 10 cm.
The EUT received DC12V charging power from the travel charger which received power through a LISN supplying power of AC 230V/50Hz.

3.3. EMI Test Receiver Setup

The EMI test receiver was set to investigate the spectrum from 150 kHz to 30MHz.
During the conducted emission

Frequency Range	IFBW
150 kHz – 30 MHz	9 kHz

3.4. Test Procedure

Power on the EUT, the EUT begins to work. Make sure the EUT operates normally during the test.
Maximizing procedure was performed on the six (6) highest emissions of the EUT.
All data was recorded in the Quasi-peak and average detection mode

3.5. Test Data

The EUT is without AC input function and therefore the test items unnecessary to test.

4. RADIATED DISTURBANCE

4.1. Radiated Emission Limit

ETSI EN 301 489-1 V2.2.3(2019-11)/EN 55032 Class B

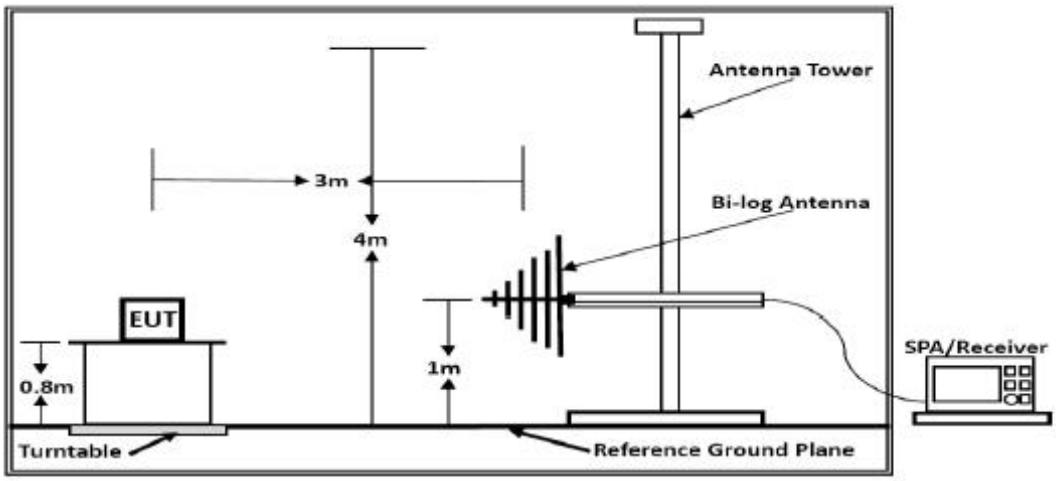
Limits for radiated disturbance Blow 1GHz

Frequency Range (MHz)	Distance (m)	Quasi-Peak limits (dB μ V/m)
30-230	10/3	30.0/40.0
230-1000	10/3	37.0/47.0
Note: (1) The smaller limit shall apply at the combination point between two frequency bands. (2) Distance refers to the distance in meters between the measuring instrument antenna and the closed point of any part of the EUT.		

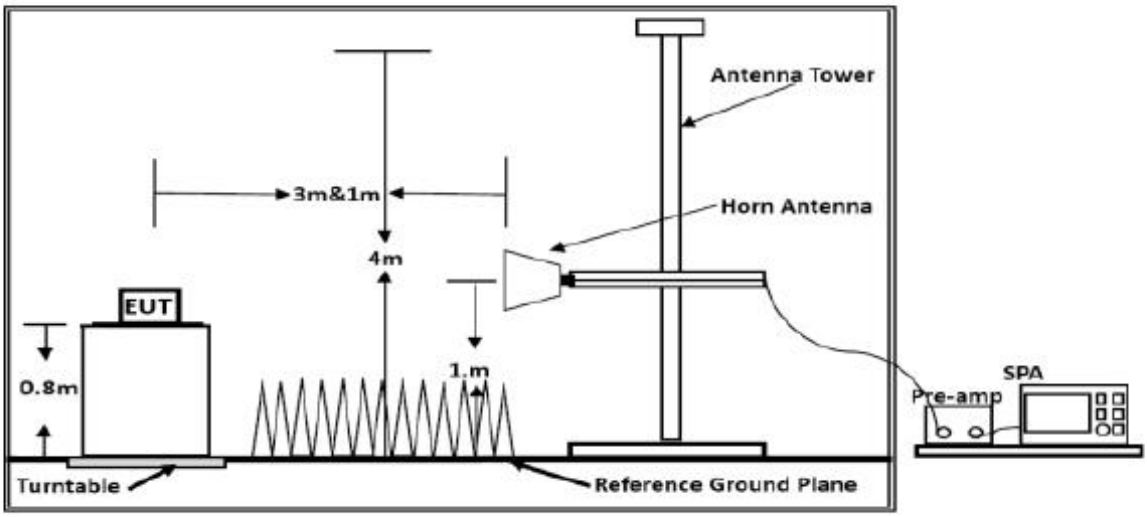
Limits for radiated disturbance Above 1GHz

Frequency Range (MHz)	Distance (m)	Quasi-Peak limits (dB μ V/m)
1000-3000	3	50 （AV）/70（PK）
3000-6000	3	54 （AV）/74（PK）
Note: (1) The smaller limit shall apply at the combination point between two frequency bands. (2) Distance refers to the distance in meters between the measuring instrument antenna and the closed point of any part of the EUT.		

4.2. Test Configuration



Below 1GHz



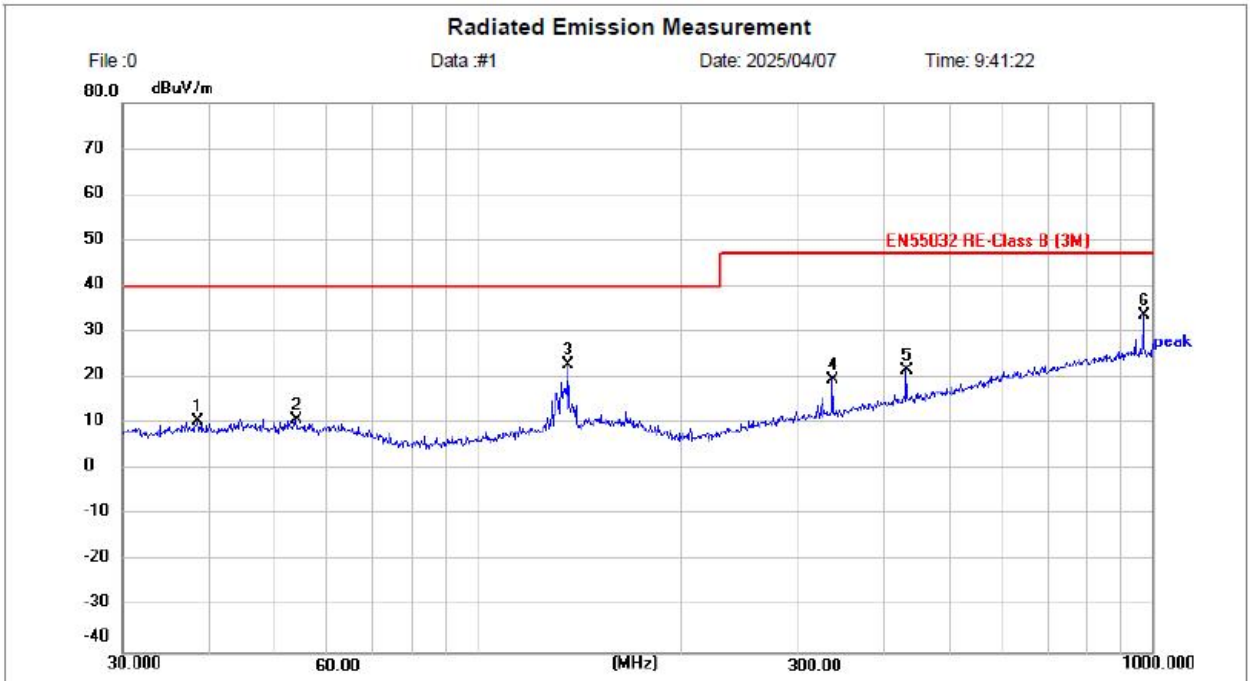
Above 1GHz

4.3. Test Procedure

Please refer to ETSI EN 301 489-1 V2.2.3(2019-11)Clause 8.2.3 and EN 55032 Annex A.2 for the measurement methods.

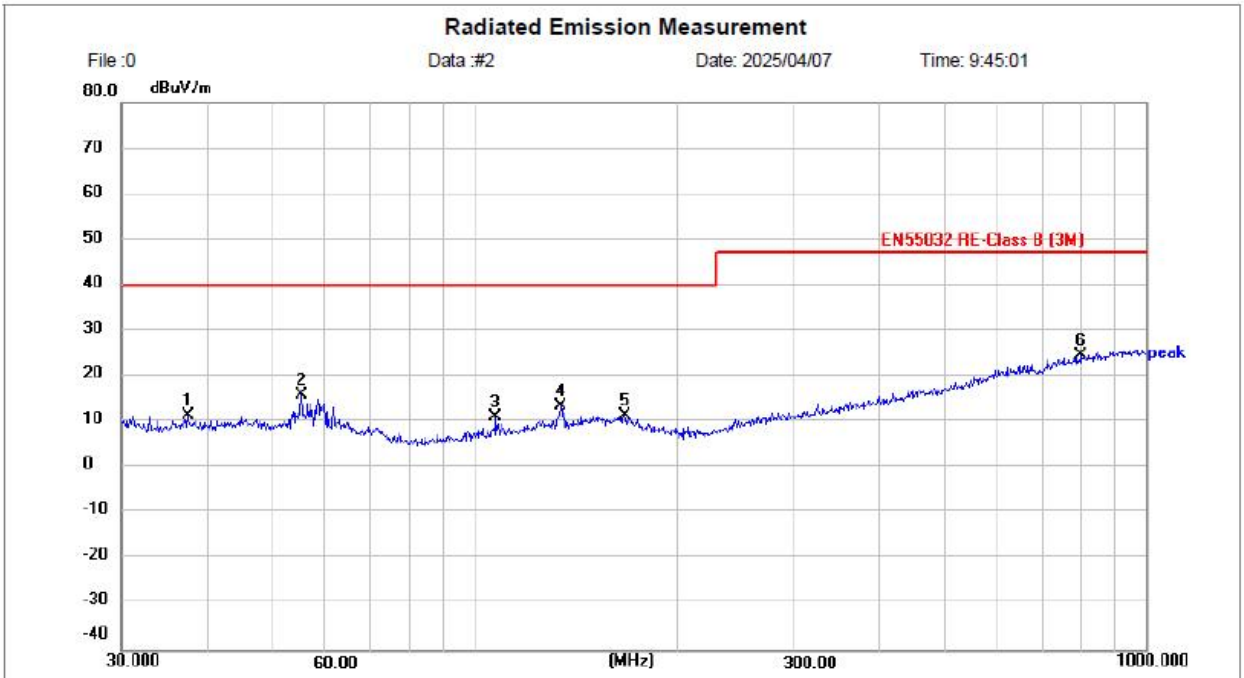
4.4. Test Data

The worst test mode of the EUT was TM1, and its test data was showed as the follow:



Site LAB
Limit: EN55032 RE-Class B (3M)
Polarization: *Horizontal*
Power:
Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	38.6160	27.06	-16.49	10.57	40.00	-29.43	peak	100	0	P	
2	53.8818	27.62	-16.73	10.89	40.00	-29.11	peak	100	0	P	
3	136.9391	39.89	-16.91	22.98	40.00	-17.02	peak	100	0	P	
4	336.0352	34.22	-14.68	19.54	47.00	-27.46	peak	100	0	P	
5	432.5457	33.67	-12.16	21.51	47.00	-25.49	peak	100	0	P	
6 *	972.3374	36.86	-3.12	33.74	47.00	-13.26	peak	100	0	P	



Site LAB
Limit: EN55032 RE-Class B (3M)
Polarization: *Vertical*
Power:
Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	37.5479	28.09	-16.67	11.42	40.00	-28.58	peak	100	360	P	
2	55.4147	32.92	-16.86	16.06	40.00	-23.94	peak	100	360	P	
3	107.8877	30.22	-19.15	11.07	40.00	-28.93	peak	100	360	P	
4	134.5592	30.46	-17.01	13.45	40.00	-26.55	peak	100	360	P	
5	167.2368	27.66	-16.12	11.54	40.00	-28.46	peak	100	360	P	
6 *	801.7863	29.12	-4.59	24.53	47.00	-22.47	peak	100	360	P	

Note: Only record the worst data.

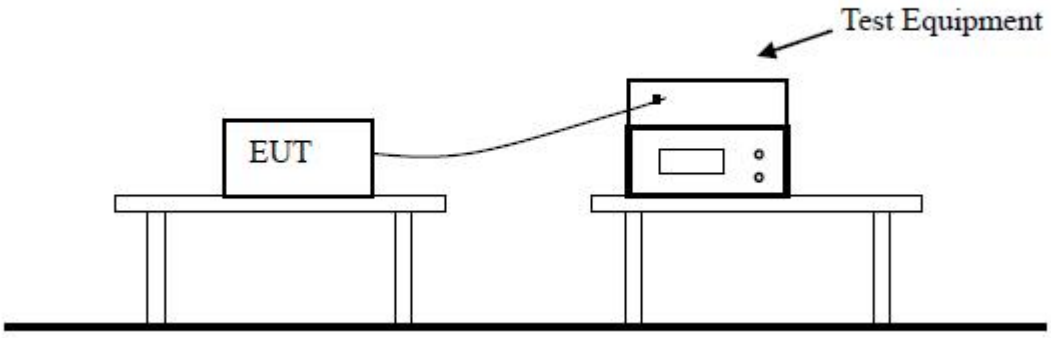
Test Mode: TM1(above 1GHz)	**Test Distance:** 3m
Test voltage:5V	**Test Results:** Passed
Detector Function: Peak+AV	

Frequency MHz	Emission Level dBμV/m		Limits dBμV/m		Margin dBμV/m		Polarization
	Peak	AV	Peak	AV	Peak	AV	
1095.18	46.86	43.68	70.00	50.00	-23.14	-6.32	H
1926.32	49.58	45.69	70.00	50.00	-20.42	-4.31	H
2154.55	48.25	40.25	70.00	50.00	-21.75	-9.75	H
3189.69	44.25	41.26	74.00	54.00	-29.75	-12.74	H
4315.45	49.69	43.69	74.00	54.00	-24.31	-10.31	H
5736.06	50.23	42.36	74.00	54.00	-23.77	-11.64	H

Frequency MHz	Emission Level dBμV/m		Limits dBμV/m		Margin dBμV/m		Polarization
	Peak	AV	Peak	AV	Peak	AV	
1237.19	47.58	45.36	70.00	50.00	-22.42	-4.64	V
1815.55	48.95	42.35	70.00	50.00	-21.05	-7.65	V
2121.46	49.68	40.25	70.00	50.00	-20.32	-9.75	V
3276.59	48.26	42.61	74.00	54.00	-25.74	-11.39	V
4321.54	49.25	43.69	74.00	54.00	-24.75	-10.31	V
5754.86	51.02	43.65	74.00	54.00	-22.98	-10.35	V

5. HARMONIC CURRENT EMISSIONS

5.1. Test Configuration



5.2. Test Standard

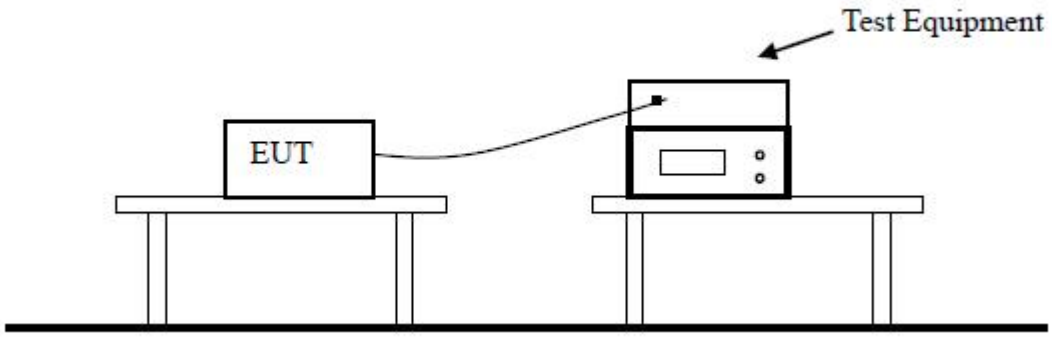
According to ETSI EN 301 489-1 V2.2.3(2019-11) & EN IEC 61000-3-2:2019+A1:2021

5.3. Test Data

Because power of EUT less than 75W, According standard EN 61000-3-2, Harmonic current unnecessary to test.

6. VOLTAGE FLUCTUATION AND FLICKER

6.1. Test Configuration



6.2. Test Standard

According to ETSI EN 301 489-1 V2.2.3(2019-11) & EN 61000-3-3: 2013/A2:2021

6.3. Test Data

Because power of EUT less than 75W, According standard EN IEC 61000-3-2, Harmonic current unnecessary to test.

7. GENERAL PERFORMANCE CRITERIA FOR IMMUNITY TEST

7.1. Performance criteria for Continuous phenomena applied to Transmitter (CT)

For equipment of type II or type III that requires a communication link that is maintained during the test, it shall be verified by appropriate means supplied by the manufacturer that the communication link is maintained during each individual exposure in the test sequence.

Where the EUT is a transmitter, tests shall be repeated with the EUT in standby mode to ensure that any unintentional transmission does not occur.

7.2. Performance criteria for Transient phenomena applied to Transmitter (TT)

For equipment of type II or type III that requires a communication link that is maintained during the test, this shall be verified by appropriate means supplied by the manufacturer during each individual exposure in the test sequence. Where the EUT is a transmitter, tests shall be repeated with the EUT in standby mode to ensure that any unintentional transmission does not occur.

7.3. Performance criteria for Continuous phenomena applied to Receiver (CR)

For equipment of type II or III that requires a communication link that is maintained during the test, it shall be verified by appropriate means supplied by the manufacturer that the communication link is maintained during each individual exposure in the test sequence. Where the EUT is a transceiver, under no circumstances shall the transmitter operate unintentionally during the test.

7.4. Performance criteria for Transient phenomena applied to Receiver (TR)

For equipment of type II or type III that requires a communication link that is maintained during the test, this shall be verified by appropriate means supplied by the manufacturer during each individual exposure in the test sequence. Where the EUT is a transceiver, under no circumstances shall the transmitter operate unintentionally during the test.

Performance criteria for ETSI EN 301 489-17 V3.3.1 (2024-09)

Criteria	During test	After test
A	Shall operate as intended. May show degradation of performance(see note 1). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance (see note 2). Shall be no loss of function. Shall be no loss of stored data or user programmable functions.
B	May show loss of function (one or more). May show degradation of performance(see note 1). No unintentional transmissions.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no degradation of performance (see note 2). Shall be no loss of stored data or user programmable functions.
C	May be loss of function (one or more).	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no degradation of performance (see note 2).

NOTE 1: Degradation of performance during the test is understood as a degradation to a level not below a minimum performance level specified by the manufacturer for the use of the apparatus as intended. In some cases the specified minimum performance level may be replaced by a permissible degradation of performance.

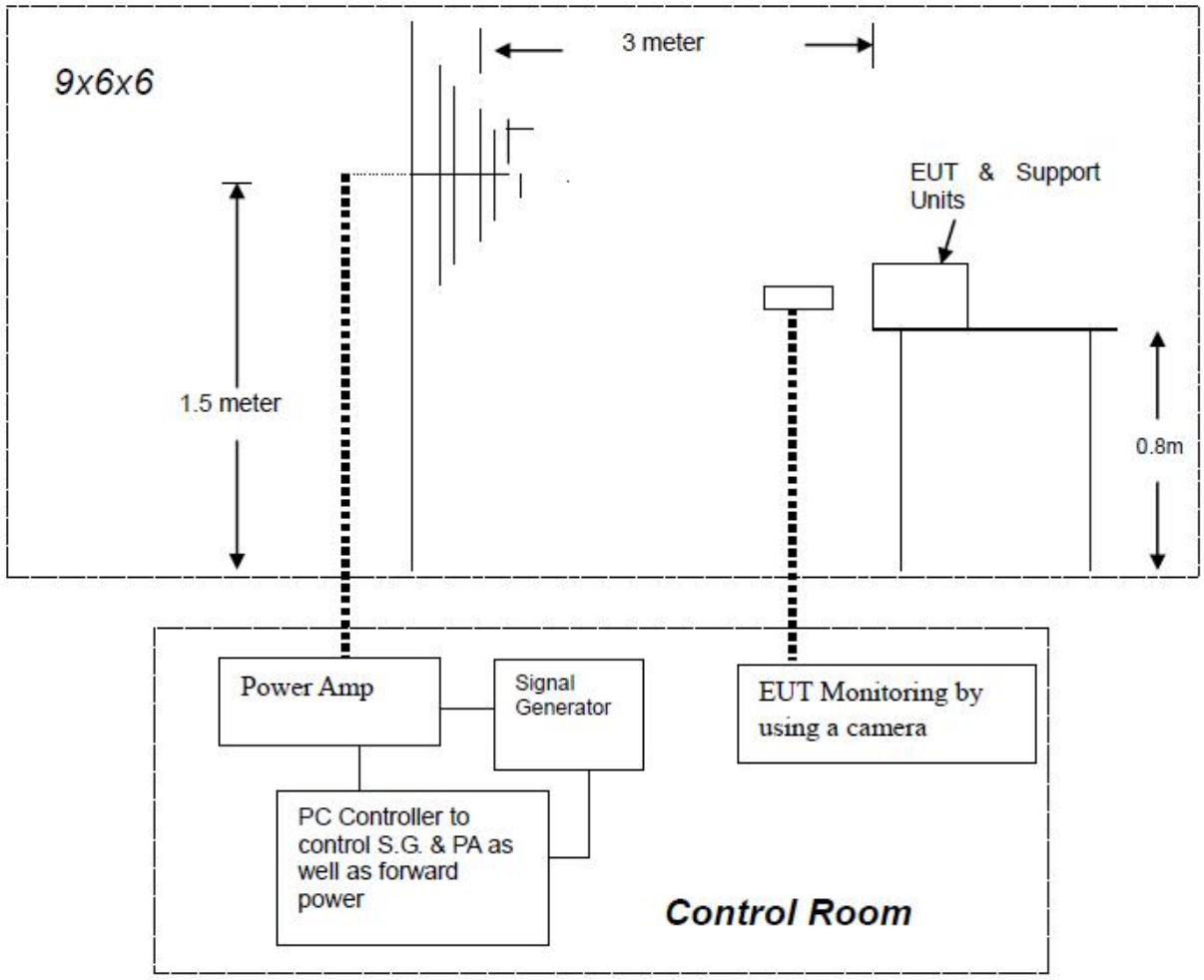
If the minimum performance level or the permissible performance degradation is not specified by the manufacturer then either of these may be derived from the product description and documentation (including leaflets and advertising) and what the user may reasonably expect from the apparatus if used as intended.

NOTE 2: No degradation of performance after the test is understood as no degradation below a minimum performance level specified by the manufacturer for the use of the apparatus as intended. In some cases the specified minimum performance level may be replaced by a permissible degradation of performance. After the test no change of actual operating data or user retrievable data is allowed.

If the minimum performance level or the permissible performance degradation is not specified by the manufacturer then either of these may be derived from the product description and documentation (including leaflets and advertising) and what the user may reasonably expect from the apparatus if used as intended.

8. RF ELECTROMAGNETIC FIELD (80 MHZ -6000 MHZ)

8.1. Test Configuration



8.2. Test Standard

ETSI 301 489-1, EN 301 489-17(EN IEC 61000-4-3:2020)
Test level 2 at 3V / m.

8.3. Severity Level

Level	Field Strength V/m
1	1
2	2
3	10
X	Special

Performance criterion: **A**

8.4. Test Procedure

The EUT and its simulators are placed on a turn table which is 0.8 meter above ground. EUT is set 3 meter away from the transmitting antenna which is mounted on an antenna tower. Both horizontal and vertical polarization of the antenna are set on test. Each of the four sides of EUT must be faced this transmitting antenna and measured individually. In order to judge the EUT performance, a CCD camera is used to monitor EUT screen. All the scanning conditions are as follows:

Condition of Test	Remarks
1. Fielded Strength	3 V/m, 80% AM @ 1kHz (Severity Level 2)
2. Radiated Signal	Unmodulated
3. Scanning Frequency	80 - 6000 MHz
4. Dwell time of radiated	0.0015 decade/s
5. Waiting Time	3 Sec.

8.5. Test Result

Test Result:

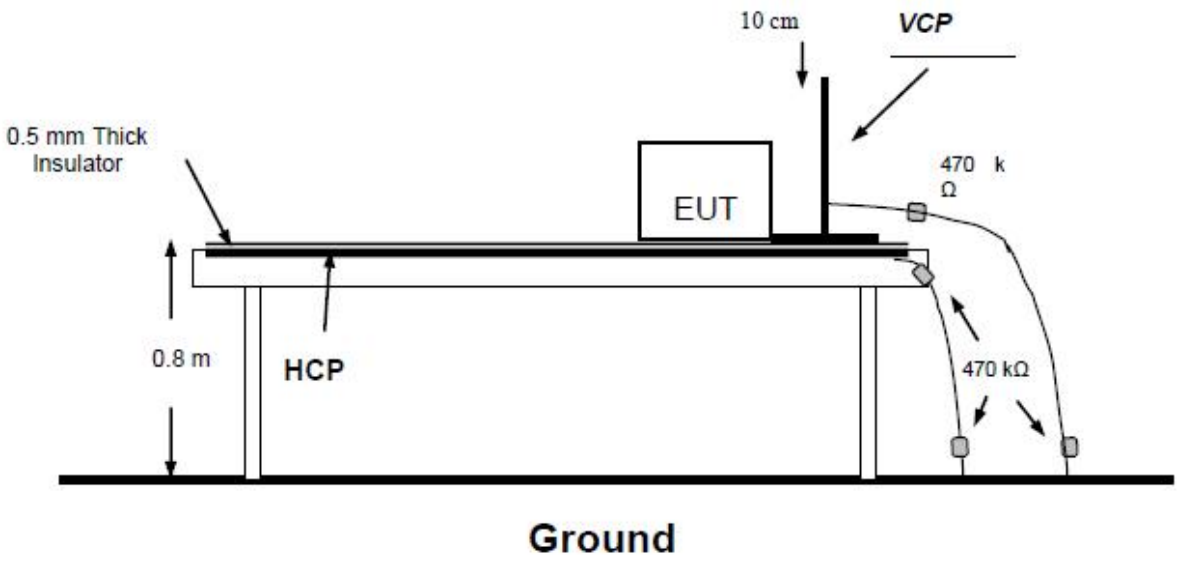
EUT Working Mode	Antenna Polarity	Frequency (MHz)	Field Strength (V/m)	Observation	Position	Conclusion
Operating Mode	Vertical	80-6000	3	CT,CR	Front, Right, Left, Back	Pass
	Horizontal	80-6000	3	CT,CR	Front, Right, Left, Back	Pass
Idle	Vertical	80-6000	3	See Note	Front, Right, Left, Back	Pass
	Horizontal	80-6000	3	See Note	Front, Right, Left, Back	Pass

***Note: Unintentional transmission is not founded from the EUT.

9. ELECTROSTATIC DISCHARGE

Please refer to ETSI EN 301 489-1 V2.2.3(2019-11)and EN 61000-4-2.

9.1. Test Configuration



EN 61000-4-2 specifies that a tabletop EUT shall be placed on a non-conducting table which is 80 centimeters above a ground reference plane and that floor mounted equipment shall be placed on a insulating support approximately 10 centimeters above a ground plane. During the tests, the EUT is positioned over a ground reference plane in conformance with this requirement.
For tabletop equipment, a 1.5 by 1.0-meter metal sheet (HCP) is placed on the table and connected to the ground plane via a metal strap with two 470 k Ohms resistors in series. The EUT and attached cables are isolated from this metal sheet by 0.5-millimeter thick insulating material. A Vertical Coupling Plane (VCP) grounded on the ground plane through the same configuration as in the HCP is used.

9.2. Test Procedure

ETSI EN 301 489-1 V2.2.3(2019-11)/ EN 61000-4-2: 2009
Test level 3 for Air Discharge at ±8 kV
Test level 2 for Contact Discharge at ±4 kV

9.2.1. Air Discharge

This test is done on a non-conductive surface. The round discharge tip of the discharge electrode shall be approached as fast as possible to touch the EUT. After each discharge, the discharge electrode shall be removed from the EUT. The generator is then re-triggered for a new single discharge and repeated 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

9.2.2. Contact Discharge

All the procedure shall be same as Section 6.2.1. except that the tip of the discharge electrode shall touch the EUT before the discharge switch is operated.

9.2.3. Indirect Discharge For Horizontal Coupling Plane

At least 10 single discharges (in the most sensitive polarity) shall be applied at the front edge of each HCP opposite the center point of each unit (if applicable) of the EUT and 0.1m from the front of the EUT. The long axis of the discharge electrode shall be in the plane of the HCP and perpendicular to its front edge during the discharge.

9.2.4. Indirect Discharge For Vertical Coupling Plane

At least 10 single discharges (in the most sensitive polarity) shall be applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, is placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges shall be applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

9.3. Test Data

PASS.

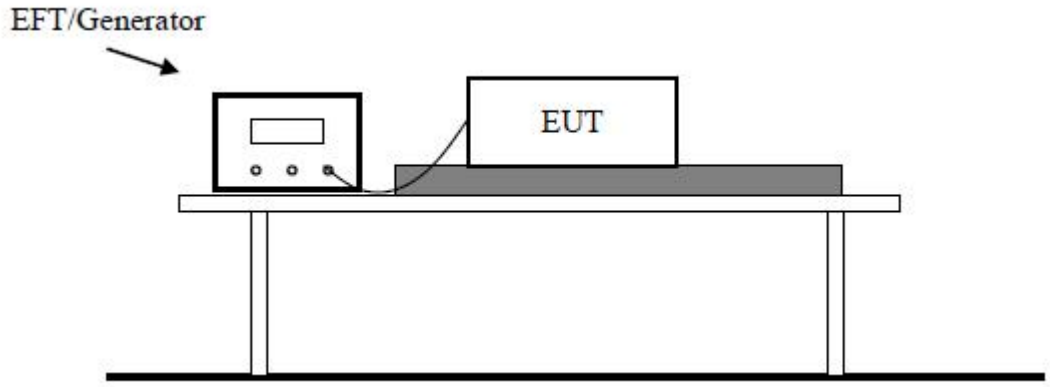
Please refer to the following page.

TEST RESULT			
Test Voltage	**Coupling**	**Observation**	**Result (Pass/Fail)**
±2KV, ±4kV	Contact Discharge	TT, TR	N/A
±2KV, ±4kV, ±8kV	Air Discharge	TT, TR	N/A
±2KV, ±4kV	Indirect Discharge HCP	TT, TR	N/A
±2KV, ±4kV	Indirect Discharge VCP	TT, TR	N/A

Note: The EUT performance complied with performance criteria for CT&CR to MS Function and there is no any degradation of performance and function.

10. ELECTRICAL FAST TRANSIENT IMMUNITY

10.1. Test Configuration



10.2. Test Standard

ETSI EN 301 489-1 V2.2.3(2019-11)/ EN61000-4-4: 2012

Test level 2 at 1 kV

Test level

Open Circuit Output Test Voltage ±10%		
Level	On Power Supply Lines	On I/O (Input/Output) Signal data and control lines
1	0.5 kV	0.25 kV
2	1 kV	0.5 kV
3	2 kV	1 kV
4	4 kV	2 kV
X	Special	Special

Performance criterion: B

10.3. Test Procedure

The EUT is put on the table, which is 0.8 meter high above the ground. This reference ground plane shall project beyond the EUT by at least 0.1m on all sides and the minimum distance between EUT and all other conductive structure, except the ground plane beneath the EUT, shall be more than 0.5m.

10.4.1. For input and output AC power ports:

The EUT is connected to the power mains by using a coupling device, which couples the EFT interference signal to AC power lines. Both polarities of the test voltage should be applied during compliance test and the duration of the test is 2 minutes.

10.4.2. For signal lines and control lines ports: No I/O ports. It's unnecessary to test.

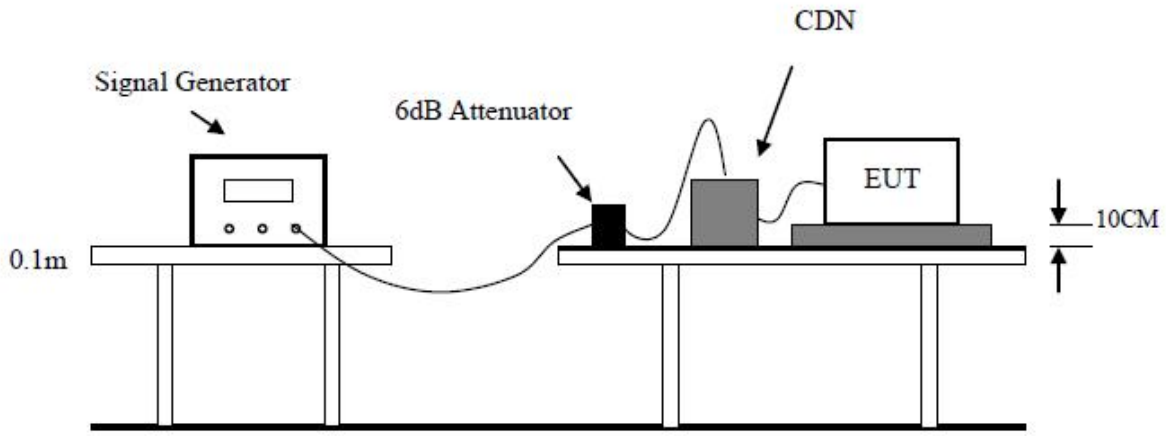
10.4.3. For DC output line ports: It's unnecessary to test.

10.4. Test Data

TEST RESULT				
Line	**Test Voltage**	**Polarity**	**Observation**	**Result (Pass/Fail)**
L	1KV	+/-	TT, TR	N/A
N	1KV	+/-	TT, TR	N/A
L-N	1KV	+/-	TT, TR	N/A

11. RF COMMON MODE

11.1. Test Configuration



11.2. Test Standard

ETSI EN 301 489-1 V2.2.3(2019-11)/ EN 61000-4-6: 2014/AC:2015
Test level 2 at 3 V (r.m.s.), 0.15 MHz ~ 80 MHz,
Modulation type: AM
Modulation depth: 80%
Modulation signal: 1 kHz

Test level

Level	Voltage Level (r.m.s..) (V)
1	1
2	3
3	10
X	Special

Performance criterion: A

11.3. Test Procedure

11.3.1. Let the EUT work in test mode and test it.

11.3.2. The EUT are placed on an insulating support 0.1 m high above a ground reference plane. CDN (coupling and decoupling device) is placed on the ground plane about 0.3 m from EUT. Cables between CDN and EUT are as short as possible, and their height above the ground reference plane shall be between 30 and 50 mm (where possible).

11.3.3. The disturbance signal described below is injected to EUT through CDN.

11.3.4. The EUT operates within its operational mode(s) under intended climatic conditions after power on.

11.3.5. The frequency range is swept from 150 kHz to 80 MHz using 3V signal level, and with the disturbance signal 80% amplitude modulated with a 1kHz sine wave.

11.3.6. The rate of sweep shall not exceed 1.5*10-3 decades/s. Where the frequency is swept incrementally, the step size shall not exceed 1% of the start and thereafter 1% of the preceding frequency value.

11.3.7. Recording the EUT operating situation during compliance testing and decide the EUT immunity criterion.

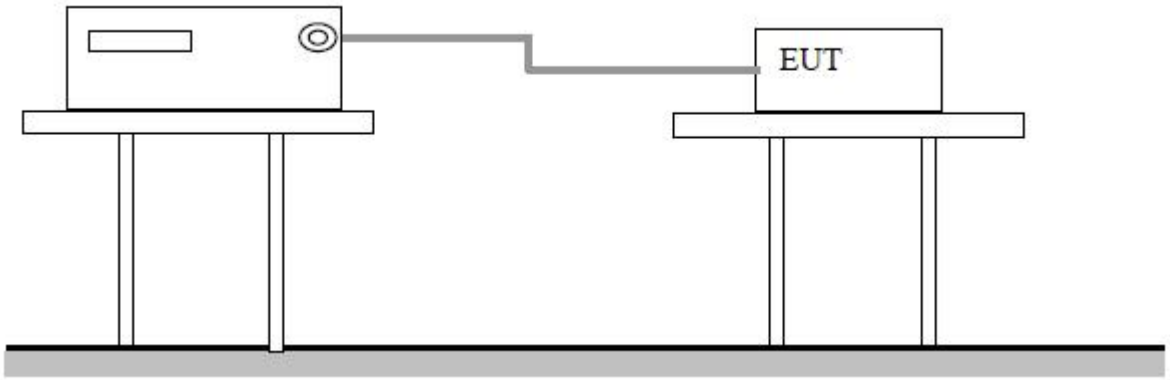
11.4. Test Data

TEST RESULT				
Frequency Range (MHz)	**Injected Position**	**Strength (Unmodulated)**	**Observation**	**Result (Pass/Fail)**
0.15 ~ 80	AC Mains	3V	CT, CR	N/A

Remark:
1. Modulation Signal:1kHz 80% AM
2. Measurement Equipment :
Simulator: CIT-10 (FRANKONIA)
CDN : ☑CDN-M2 (FRANKONIA)
☐CDN-M3 (FRANKONIA)

12. SURGES, LINE TO LINE AND LINE TO GROUND

12.1. Test Configuration



12.2. Test Standard

ETSI EN 301 489-1 V2.2.3(2019-11)/ EN 61000-4-5: 2014+A1:2017

L-N: Test level 2 at 1 kV

L-PE, N-PE Test Level 3 at 2Kv

Test Level

Open Circuit Output Test Voltage ±10%		
Level	On Power Supply Lines	On I/O (Input/Output) Signal data and control lines
1	0.5 kV	0.25 kV
2	1 kV	0.5 kV
3	2 kV	1 kV
4	4 kV	2 kV
X	Special	Special

Performance criterion: B

12.3. Test Procedure

12.3.1. For line to line coupling mode, provide a 0.5 kV 1.2/50us voltage surge (at open-circuit condition).

12.3.2. At least 5 positive and 5 negative (polarity) tests with a maximum 1/min repetition rate are conducted during test.

12.3.3. Different phase angles are done individually.

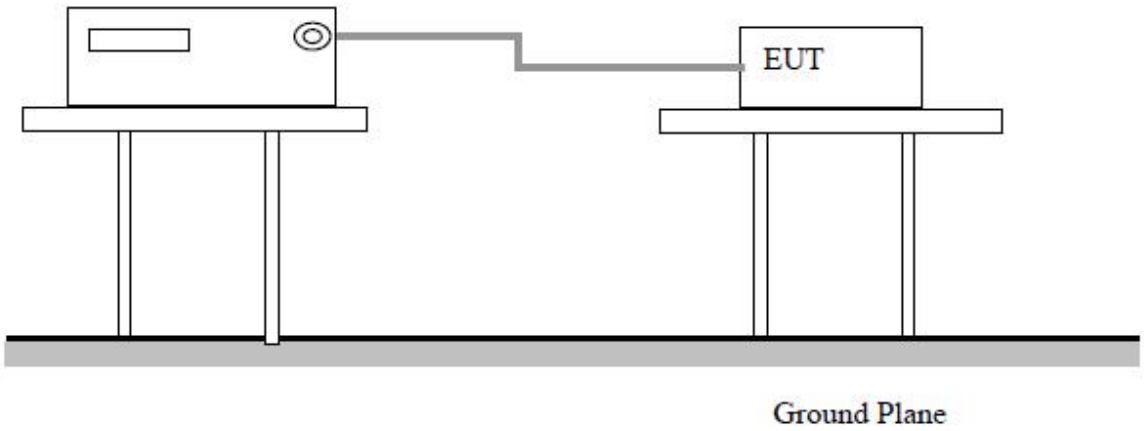
12.3.4. Record the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.

12.4. Test Data

TEST RESULT						
Location	**Polarity**	**Phase Angle**	**Number of Pulse**	**Pulse Voltage (KV)**	**Observation**	**Result (Pass/Fail)**
L-N	+	0°, 90°, 180°, 270°	5	1.0	TT, TR	N/A
	-	0°, 90°, 180°, 270°	5	1.0	TT, TR	N/A

13. VOLTAGE DIPS/INTERRUPTIONS IMMUNITY TEST

13.1. Test Configuration



13.2. Test Standard

ETSI EN 301 489-1 V2.2.3(2019-11)/ EN IEC 61000-4-11: 2020
Test levels and Performance Criterion

Test Level

Voltage Reduction %UT	Voltage dips %UT	Duration (in period)
100	0	0.5
100	0	1
30	70	5
Voltage Reduction %UT	Voltage Interruptions %UT	Duration (in period)
100	0	250

Performance criterion: B&C

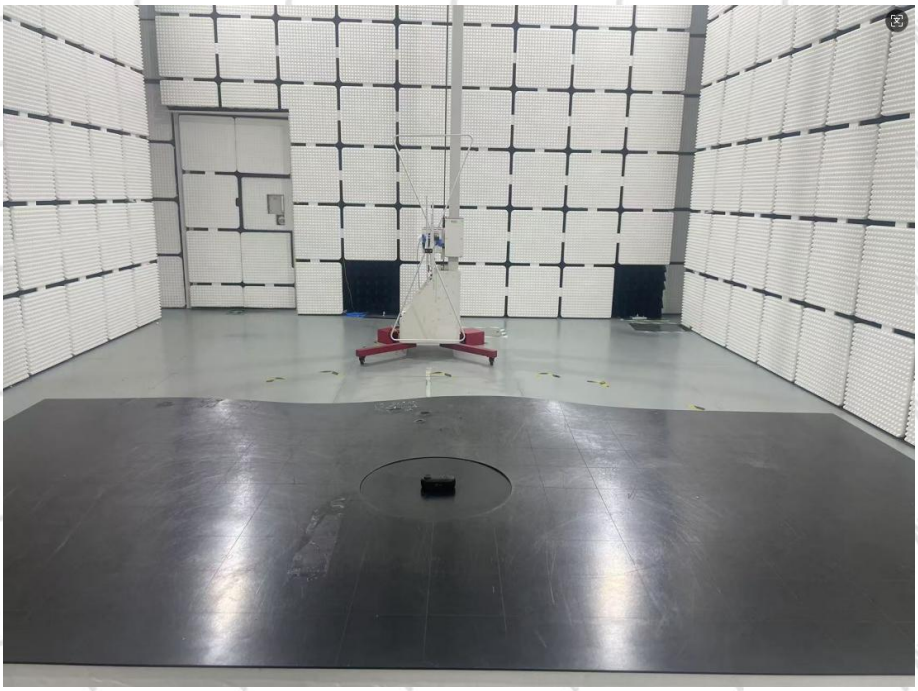
13.3. Test Procedure

13.3.1. The interruption is introduced at selected phase angles with specified duration.
13.3.2. Record any degradation of performance.

13.4. Test Data

TEST RESULT				
Test Level % U_T	**Voltage Dips & Short Interruptions % U_T**	**Duration (in periods)**	**Observation**	**Result (Pass/Fail)**
0	100	0.5P	TT, TR	N/A
0	100	1P	TT, TR	N/A
70	30	25P	TT, TR	N/A
0	100	250P	TT, TR	N/A

14. TEST SETUP PHOTOGRAPHS





15. EUT EXTERIOR AND INTERIOR PHOTOGRAPHS





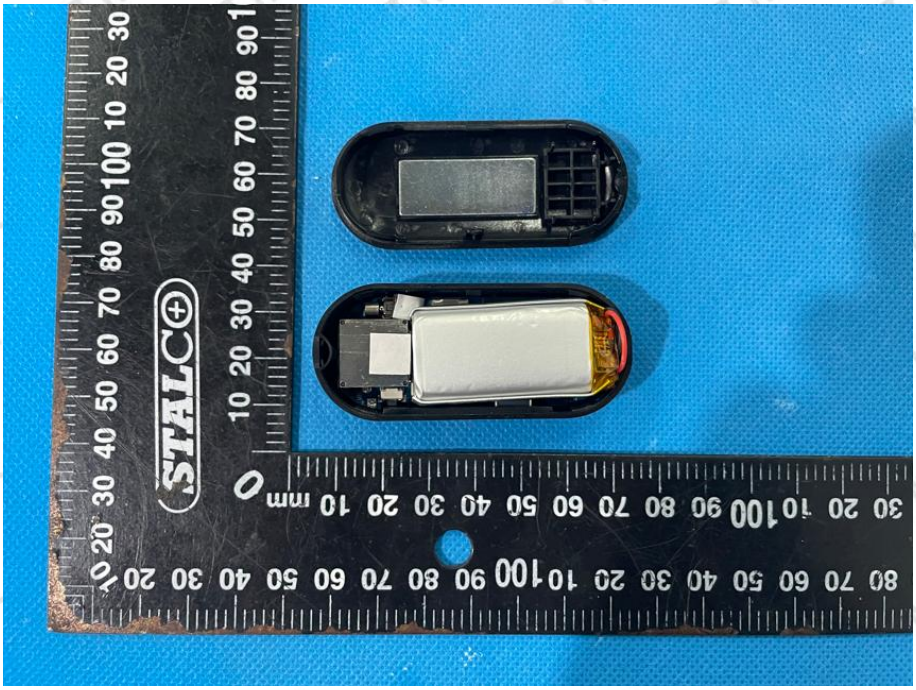

UHD Pro
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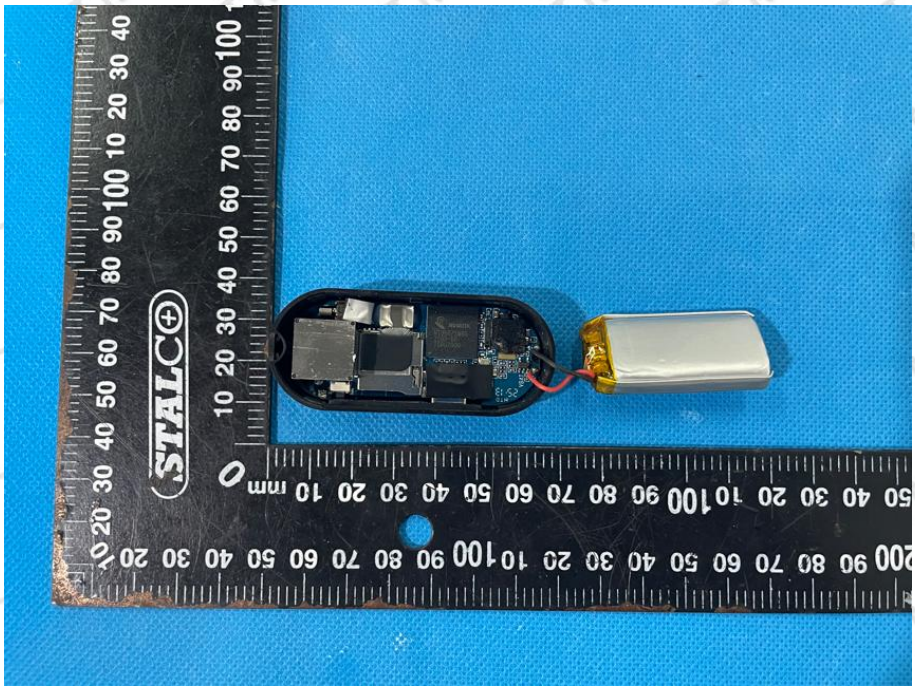

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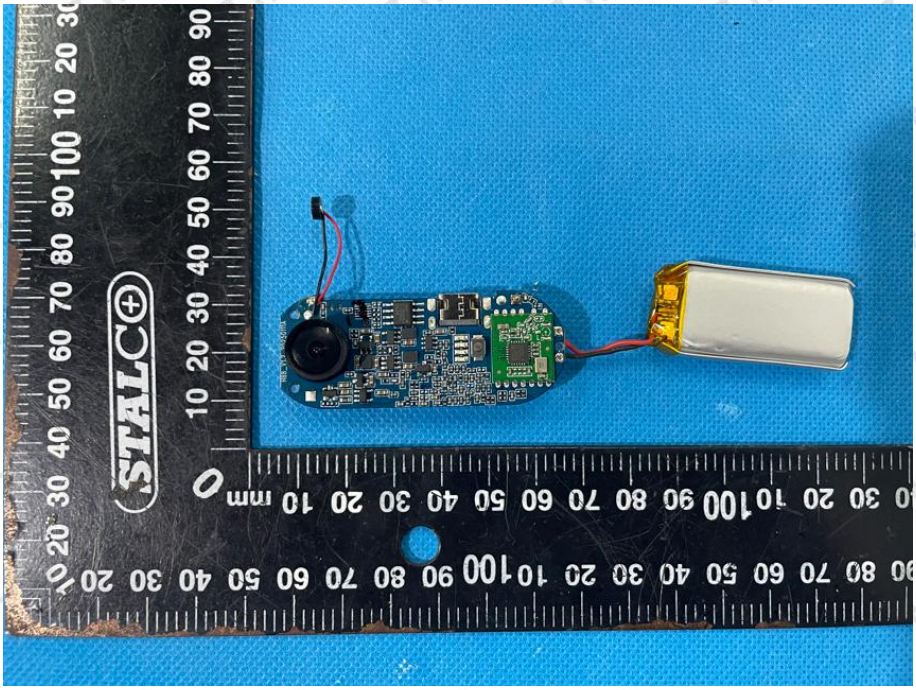

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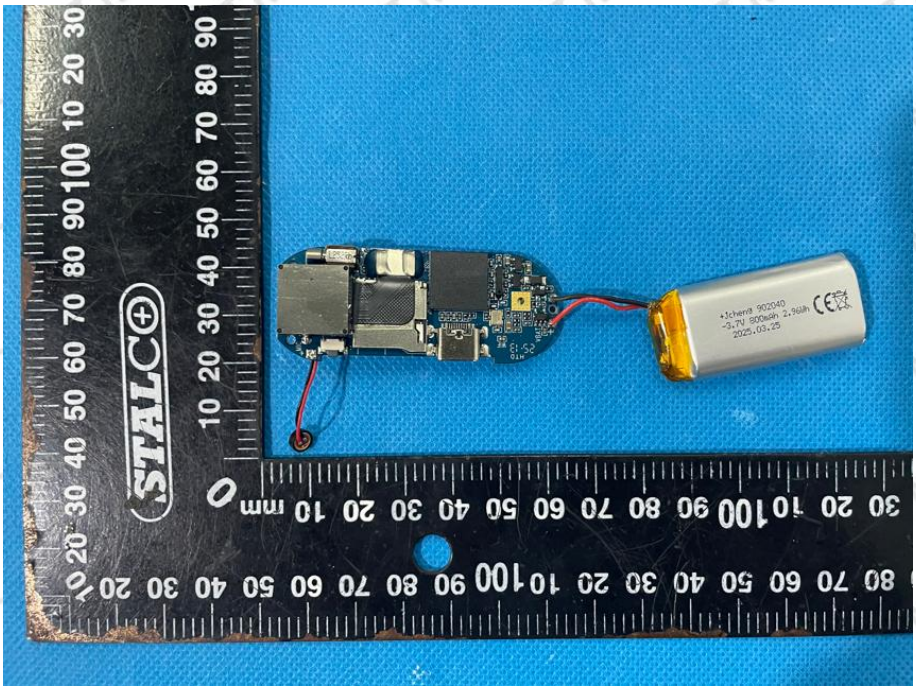

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16.	Test Equipments				
Instrument Type	Manufacturer	Model	Serial No.	Date of Cal.	Due Date
ESPI Test Receiver	ROHDE&SCHWARZ	ESPI 3	100379	2024-6-18	2025-6-17
Absorbing Clamp	ROHDE&SCHWARZ	MDS-21	100126	2024-6-18	2025-6-17
Electrostatic analog generator	LIONCEL	ESD-203B	0210502	2024-6-18	2025-6-17
Signal Generator	HP	8648A	3633A02081	2024-6-18	2025-6-17
Amplifier	A&R	500A100	17034	2024-6-18	2025-6-17
Amplifier	A&R	100W/1000M1	17028	2024-6-18	2025-6-17
Isotropic Field Monitor	A&R	FM2000	16829	2024-6-18	2025-6-17
Isotropic Field Probe	A&R	FLW220100	16755	2024-6-18	2025-6-17
Biconic Antenna	EMCO	EVOD PROTANK8	9507-2534	2024-6-18	2025-6-17
Log-periodic Antenna	A&R	AT1080	16812	2024-6-18	2025-6-17
Injection Clamp	EMTEST	F-2031-23MM	368	2024-6-18	2025-6-17
Attenuator	EMTEST	ATT6	0010222a	2024-6-18	2025-6-17
Computer	IBM	8434	1S8434KCE99BLXLO*	-	-
Oscillator	KENWOOD	AG-203D	3070002	2024-6-18	2025-6-17
Spectrum Analyzer	HAMEG	HM5012	-	-	-
Power Supply	LW	APS1502	-	-	-
5K VA AC Power Source	California Instruments	5001iX	56060	2024-6-18	2025-6-17
CDN	EM TEST	CDN M2/M3	-	2024-6-18	2025-6-17
Attenuation	EM TEST	ATT6/75	-	2024-6-18	2025-6-17
Resistance	EM TEST	R100	-	2024-6-18	2025-6-17
Electromagnetic Injection Clamp	LITTHI	EM101	35708	2024-6-18	2025-6-17
Inductive Components	EM TEST	MC2630	-	2024-6-18	2025-6-17
Antenna	EM TEST	MS100	-	2024-6-18	2025-6-17
Signal Generator	ROHDE&SCHWARZ	SMT03	100029	2024-6-18	2025-6-17
Power DJ MIXER	AR	150W1000	300999	2024-6-18	2025-6-17
Field probe	Holaday	HI-6005	105152	2024-6-18	2025-6-17
Bilog Antenna	Chase	CBL6111C	2576	2024-6-18	2025-6-17
Loop Antenna	EMCO	6502	00042960	2024-6-18	2025-6-17

ESPI Test Receiver	ROHDE&SCHWARZ	ESI7	838786/013	2024-6-18	2025-6-17
3m OATS	--	--	N/A	2024-6-18	2025-6-17
Horn Antenna	SCHWARZBECK	VULB9168	N/A	2024-6-18	2025-6-17
Horn Antenna	SCHWARZBECK	BBHA9120D	N/A	2024-6-18	2025-6-17
Power meter	Anritsu	ML2487A	6K00003613	2024-6-18	2025-6-17
Power sensor	Anritsu	MA2491A	32263	2024-6-18	2025-6-17
Bilog Antenna	Schwarebeck	VULB9163	9163/340	2024-6-18	2025-6-17
9*6*6 Anechoic	--	--	N/A	2021-08-21	2025-8-20
Test Receiver	Rohde&Schwarz	ESC17(9kHz-7GHz)	100336	2024-6-18	2025-6-17
Broadband antenna	Schwarzbeck	VULB9168	01222	2024-6-18	2025-6-17
Horn antenna	Schwarzbeck	BBHA9120D	02476	2024-6-18	2025-6-17
Preamplifier	Schwarzbeck	BBV9745	00250	2024-6-18	2025-6-17
Preamplifier	N/A	TRLA-01018G440B	21081001	2024-6-18	2025-6-17
3M method semi anechoic chamber	SKET	9m*6m*6m	2021082304	2021-8-23	2025-8-22
Pointer hygrometer	M&G	ARC92570	N/A	2024-6-18	2025-6-17

End of the report